Visium spatial transcriptomics data analysis with Chipster 2025

PART I: Working with 1 slice

In this tutorial we analyze spatially resolved transcriptomics data generated with the Visium technology from 10x Genomics. The data contains two sagittal mouse brain slices generated using the Visium v1 chemistry.

We have downloaded the data (output files of the Visium Space Ranger pipeline) from here and imported in Chipster two sets of files for you: one set for an anterior section and one for a matched posterior section. We also imported a reference scRNA-seq dataset of ~14,000 adult mouse cortical cell taxonomy from the Allen Institute, generated with the SMART-Seq2 protocol.

Open Chipster: Go to **https://chipster.csc.fi/,** click on **Launch Chipster**, and log in.

1. Open training session
Click **Sessions** and select **course_spatial_transcriptomics_visium_2025** under **Training sessions**. Rename the session course_spatial**_your_first_name**

2. Find the right files and make a tar package of them
Go to the tool category **Spatially resolved transcriptomics (Seurat v5)** and select the tool **Seurat v5 -Setup and QC.** Open the manual by clicking the **More info** link. Check out the names of the input files needed. Select these 5 input files (Space Ranger files) for the anterior sample, and the tool **Utilities / Make a .tar package**. Name your package as **mouse_anterior** and **run** the tool by selecting **Run Tool (1 job)**.

3. Set up the Seurat object and perform quality control
Select the **mouse_anterior.tar** package generated in the previous step and the tool **Spatially resolved transcriptomics (Seurat v5) / Seurat v5 -Setup and QC.** Check the parameters and set **Name of the sample** = **anterior1**. **Run** the tool.
Select the **QC_plots.pdf** and click **Open in new tab**. Look at all the pages.

-What does a spot in the image represent?
-Can you see any spatial variation in the number of transcripts?
-In which region do you see high mitochondrial transcript percentage? What can it mean?
-What is high hemoglobin percentage signalling?
-What would be the optimal limits for the mitochondrial transcript percentage (percent.mt) and hemoglobin transcript percentage (percent.hb)?

4. Filter spots
Select **seurat_spatial_setup_obj.Robj** and the tool **Seurat v5 -Filter spots.** Would the default spot filtering parameters be good for this dataset, based on the QC plots? In order to follow the Seurat vignette, let's not remove any spots for now: set the mitochondrial and hemoglobin filtering parameters to **100**, and the ribosomal parameter to **0**, and run the tool.

5. Normalize data and detect highly variable genes with SCTransform, and perform principal component analysis
Select **seurat_obj_filtered.Robj** and run the tool **Seurat v5 -Normalization and PCA** using the default parameter values. Look at the plots in the **PCAplots.pdf**.

-What should we look for in the elbow plot?
-Based on the elbow plot, how many PCs could be appropriate for the next step?
-What do the other two visualizations tell us?

6. Clustering
Select the **seurat_spatial_obj_pca.Robj** and the tool **Seurat v5 -Clustering**. Based on the elbow plot, choose **30** as the **Dimensions of reduction to use** parameter that sets the number of PCs used for this clustering step. Use **0.8** as the **Resolution for granularity for clustering** parameter. Open the **clustering_plots.pdf** in a new tab.

-How many clusters are there? Does the coloring (= clustering) match the grouping found by UMAP?
-How can we adjust the number of clusters that are produced?
-Are the clusters located in particular areas of the tissue?
-Cerebral cortex, also called gray matter, is the brain's outermost layer of nerve cell tissue. Which clusters correspond to the cortex?

7. Gene expression visualization
Select the **seurat_obj_clustering.Robj** from the previous step and run the tool **Seurat v5 - Visualize gene expression** with the default parameters for genes **Hpca** and **Ttr**. Then run the tool again but change **Minimum transparency = 0.1**. Open the resulting **Feature_plot.pdf** files in new tabs and compare them.

-Are these two genes spatially variable based on these plots?
-What is the difference between these two plots?

8. Visualize clusters
Looking at the colorful spatial cluster plots can give you a migraine. To more easily detect where the clusters are located, let's visualize some of them one by one. Select **seurat_obj_clustering.Robj** and the tool **Seurat v5 -Visualize clusters**. In the parameters, type **clusters to plot = 1, 2, 3, 4, 5, 6, 7, 8.** Open the resulting **spatialdimplot.pdf.**

-Which of the clusters correspond to the outermost layer? Does it have any spots elsewhere in the tissue?
-Can you spot the cortex clusters more easily now?

9. Identify spatially variable genes based on clusters
Choose **seurat_obj_clustering.Robj** and select the tool **Seurat v5 -Identify spatially variable genes based on clusters.** Type clusters to compare to the parameter fields: **first cluster = 5** and **second cluster = 6** and run the tool. Open the resulting **Markerplot.pdf** and compare it to the **spatialdimplot.pdf** for clusters 5 and 6.

-What are the 3 most spatially variable genes between these two clusters?

Now run the tool again for the clusters **5** and **6** but change the parameter **Determine color scale based on all genes** to **yes.** Compare the resulting **Markerplot.pdf** to the previous **Markerplot.pdf.**

-What happened to the color scales in the new **Markerplot.pdf**? Does this make it easier to compare the gene expression between the genes?
-Based on this new **Markerplot.pdf**, where can you see highest gene expression values?

Now open either of the **spatially_variable_genes.tsv** files.

-Which cluster is expressing which of these three genes? Are there differences in levels (see columns pct.1, pct.2 and avg_log2FC)?

10. Identify spatially variable genes without pre-annotation
Previous step compared two clusters. With the tool **Seurat v5 -Identify spatially**

variable genes without pre-annotation, we can identify genes that have spatial patterning without taking cluster information into account. Run this tool using the default parameter values. Open the file **Markerplot2.pdf**.

-What are the 6 most spatially variable genes?
-Look at the color scales. Could it be easier to compare the gene expression values between the genes if the parameter **Determine color scale based on all genes** was set to **yes?**

BONUS HOMEWORK: We have now run the tool using the Moran's I method, but you can also run the tool using the Mark Variogram method and compare the identified spatially variable genes.

<u>11. Subset anatomical regions</u>
Let's approximately subset the frontal cortex spots. Choose **seurat_obj_clustering.Robj** and the tool **Seurat v5 - Subset out anatomical regions based on clusters.** Type **subset of clusters = 3, 4, 6, 7** and run the tool. Open **subset.pdf**.

-Did you approximately get the cortex spots?

<u>12. Annotate spots by integrating spatial data with a scRNA-seq reference data</u>
Select the **seurat_obj_subset.Robj** generated in the previous step. Now that we have subsetted the original data, we need to renormalize and run PCA for this subsetted object using the tool **Seurat v5 -Normalization and PCA** and using the default parameter values.
After running the renormalization and PCA for the subset, you can choose both the **seurat_spatial_obj_pca.Robj** and the **allen_cortex.rds** reference file already uploaded in the session, and select the tool **Seurat v5 -Integration with scRNA-seq data**. Set the parameter **Number of subsampling cells in SCTransform** = 3000 and check that the input files are correctly assigned. Run the tool (it takes some time). Open the resulting **reference_UMAP_plot.pdf**.
-What do you see in the plot? What does each dot represent?

Select the **seurat_obj_integrated.Robj** and run the tool **Seurat v5 -Visualize cell types after integration with scRNA-seq data** using the default parameters. Open **integration_plot.pdf**.

- Where are those spots predicted to contain L2/3 IT cells located in our subsetted data?
-What are the top identified spatially variable cell types in our subsetted data?

PART II: Joint analysis of two samples

1. Prepare the other slice

Repeat steps 2, 3 and 4 for the other sample (with same filtering parameters). Name the tar-package as **mouse_posterior,** and the sample as **posterior1**. Open **QC_plots.pdf**.

-How does this sample look? Does it look very different?

2. Perform normalization, combine samples and run PCA

Choose the two **seurat_obj_filtered.Robj** objects and the tool **Seurat v5 –Combine multiple samples, normalization and PCA**. As the combining method, choose **Merge**. Run the tool. While the tool is running, study the manual of this tool and answer these questions:

-What are the two different combining methods, and how do they differ?
-Why do you think we used "merge" here?

Look at the produced **logfile.txt**.

-What does the **logfile.txt** tell us? Did we use the selected 3000 as the **Number of variable genes in combined object** for running PCA?
-What should we try to do, if we want to use the selected 3000?

BONUS exercise: You can try this at home later. Choose **5000** as the **Number of variable genes to return in SCTransform**. Is a **logfile.txt** produced now and how many variable genes are used for running PCA?

3. PCA, clustering, and visualization for combined samples

Choose the **seurat_obj_multiple.Robj** from the previous step and run the tool **Seurat v5 - Clustering** for this combined object using the default parameters. Open the **UMAP_plot.pdf**.

-How many clusters are there in this merged object?
-Are the clusters present in both samples?

4. Visualize gene expression of two samples

Choose **seurat_obj_clustering.Robj** and tool **Seurat v5 -Visualize gene expression**. Type **gene name(s) = Hpca, Plp1** and run the tool. Open the resulting **Feature_plot.pdf**.
-Does it seem that the expression of these genes follows a pattern between the two images?

5. BONUS exercise: Repeat the analysis with "integrate" instead of "merge"

Repeat steps 2-4, but now in step 2, choose "integrate" as the combining method instead. Compare the clusters from merged and integrated analysis. Can you see any differences? When comparing the results you can use the Allen brain atlas as a reference for brain regions.

6. Send a support request to the Chipster team

In the top panel, click **Contact**. Click the **Contact support** button. In the new window,

- -Click **Attach a copy of your last session** XXX
- -Enter your **email address** and write a small **message.**

7. Share a session with a colleague (in this case with Eija and Maria)

Go to the **Session info** panel, click the **three dots** next to the session name, and select **Share**.

- -click **Add rule**.
- -In the **UserID** field, enter **jaas/support_session_owner**
- -set **Rights** = **Read-only** (you don't want us to mess up your session)
- -Click **Save** and **Close**.

Check what is your own UserID: Click on your **username** (top right corner) and select **Account**.